

Laboratory Exercise

Design a VPC

Objectives:

At the end of the exercise, the students should be able to:

- Create a VPC architecture diagram using AWS architecture components.
- Illustrate the use of public and private subnets across multiple Availability Zones.

Requirements:

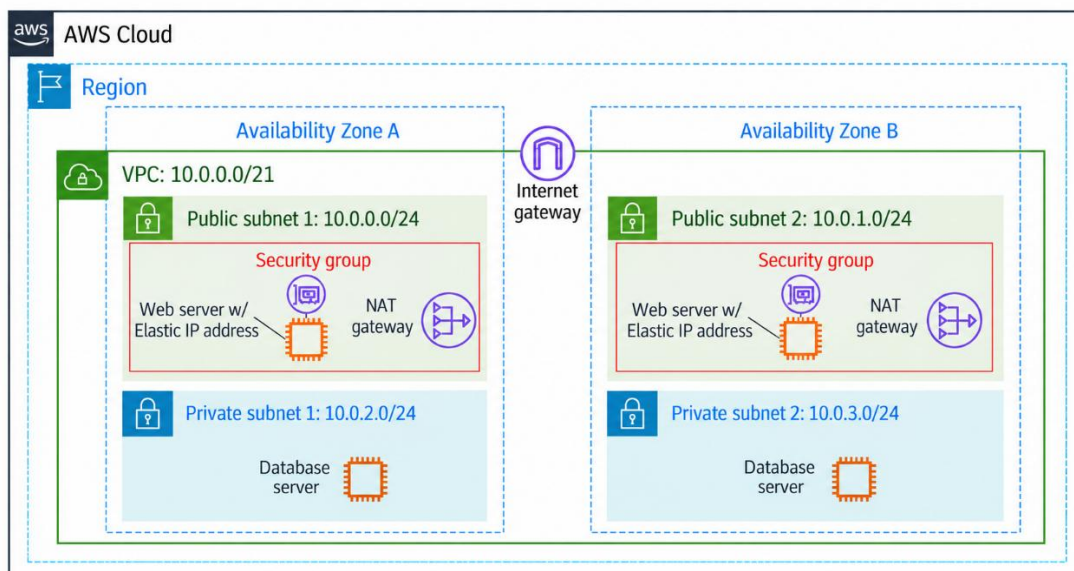
- Web browser
- <https://app.diagrams.net/>
- MS Word

Procedure:

1. Open <https://app.diagrams.net/> (Draw.io) using a web browser.
2. Click **More Shapes** at the bottom of the Shapes panel. Select **AWS 2026**, then click **Apply**.
3. Review the following scenario:

You have a small business with a website that is hosted on an Amazon Elastic Compute Cloud (Amazon EC2) instance. You have customer data that is stored on a backend database that you want to keep private. You want to use Amazon VPC to set up a VPC that meets the following requirements:

- Your web server and database server must be in separate subnets.
 - The first address of your network must be 10.0.0.0. Each subnet must have 256 total IPv4 addresses.
 - Your customers must always be able to access your web server.
 - Your database server must be able to access the internet to make patch updates.
 - Your architecture must be highly available and use at least one custom firewall layer.
4. Use the reference diagram below as a guide in recreating the VPC architecture in Draw.io.



5. Use the following AWS 2026 shape categories to recreate the reference diagram:
 - **AWS / Groups** – AWS Cloud, Region, VPC, Availability Zones, subnets, and security groups.
 - **AWS / Compute** – Compute icon used for the web servers and database servers.
 - **AWS / Network & Content Delivery** – Elastic Network Interface, Internet Gateway, and NAT Gateway.
6. Add text labels where necessary to identify the components, servers, security controls, and CIDR blocks shown in the reference diagram.
7. Click **File > Export as > PNG** and save the completed VPC architecture diagram.
8. Insert the exported diagram into MS Word. Include your name and section above the diagram.
9. Below the diagram, briefly explain how the completed architecture satisfies the following scenario requirements:
 - Separation of the web server and database server
 - 256 total IPv4 addresses per subnet
 - Customer access to the web server
 - Internet access for database patch updates
 - High availability and custom firewall protection
10. Show your work to your instructor before submitting a PDF copy in eLMS.

GRADING RUBRIC (40 points)

Criteria	Excellent (10)	Satisfactory (8)	Needs Improvement (6)	Points
VPC and Subnet Structure	VPC, Availability Zones, subnets, and CIDR blocks accurately follow the reference diagram.	Most structural elements follow the reference with minor errors.	Several structural elements are missing or inaccurate.	___/10
AWS Components	All required AWS components are correctly placed and labeled.	Most components are correctly placed and labeled with minor errors.	Several components are missing, misplaced, or incorrectly labeled.	___/10
Architecture Explanation	Clearly explains how all scenario requirements are addressed by the architecture.	Explains most requirements correctly with minor inaccuracies.	Explains some requirements with major inaccuracies or omissions.	___/10
Diagram Completeness and Clarity	Diagram is complete, organized, readable, and properly aligned.	Diagram is mostly complete with minor layout or labeling issues.	Diagram has several missing labels or major presentation issues.	___/10
			TOTAL SCORE	___/40